

INTERNATIONAL STANDARD



Plasma display panels – Part 4-1: Environmental testing methods – Climatic and mechanical



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PLASMA DISPLAY PANELS –**Part 4-1: Environmental testing methods –
Climatic and mechanical**

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The text of this standard is based on the following documents:

FDIS	Report on voting
110/637/FDIS	110/653/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all the parts in the IEC 61988 series, under the general title *Plasma display panels*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

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PLASMA DISPLAY PANELS –

Part 4-1: Environmental testing methods – Climatic and mechanical

1 Scope

This part of IEC 61988 defines the following test methods for evaluating the climatic and mechanical endurance characteristics of plasma display modules (PDP modules):

- a) temperature and humidity tests
- b) air pressure tests
- c) mechanical tests

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60068-1:2013, *Environmental testing – Part 1: General and guidance*

IEC 60068-2-1:2007, *Environmental testing – Part 2: Tests – Test A: Cold*

IEC 60068-2-2:2007, *Environmental testing – Part 2: Tests – Test B: Dry heat*

IEC 60068-2-6, *Environmental testing – Part 2: Tests – Test Fc: Vibration (sinusoidal)*

IEC 60068-2-13:1983, *Basic environmental testing procedures – Part 2-13: Tests – Test M: Low air pressure*

IEC 60068-2-14:2009, *Environmental testing – Part 2-14: Tests – Test N: Change of temperature*

IEC 60068-2-27:2008, *Environmental testing – Part 2-27: Tests – Test Ea and guidance: Shock*

IEC 60068-2-30:2005, *Environmental testing – Part 2-30: Tests – Test Db: Damp heat, cyclic (12 h + 12 h cycle)*

IEC 60068-2-78:2012, *Environmental testing – Part 2-78: Tests – Test Cab: Damp heat, steady state*

IEC 61747-5:1998, *Liquid crystal and solid-state display devices – Part 5: Environmental, endurance and mechanical test methods*

IEC 61988-1, *Plasma display panels – Part 1: Terminology and letter symbols*

IEC 61988-2-1, *Plasma display panels – Part 2-1: Measuring methods – Optical and optoelectrical*

IEC 61988-2-5, *Plasma display panels – Part 2-5: Measuring methods – Acoustic noise*

ISO 2248, *Packaging – Complete, filled transport packages – Vertical impact test by dropping*

ISO 4180, *Packaging – Complete, filled transport packages – General rules for the compilation of performance test schedules – Part 1: General principles*

ISO 10531, *Packaging – Complete, filled transport packages – Stability testing of unit loads*

3 Terms, definitions and letter symbols

For the purposes of this document, the terms, definitions and letter symbols given in IEC 60068-1 and IEC 61988-1, as well as the following apply.

3.1

P_{op}

air pressure at which the PDP module is operated during the tests

3.2

P_{st}

air pressure at which the PDP module is stored in a non-operating state during the tests

4 Environmental tests

4.1 General

In this document, tests mean storage or operation tests at high or low temperature, high humidity, low pressure, or combinations thereof, and vibration tests or shock tests, while measurements mean the characteristic evaluations, i.e. initial, intermediate and final measurements.

4.2 Test items

This document describes the following test items:

- a) temperature and humidity tests (operation);
- b) temperature and humidity tests (storage);
- c) air pressure tests;
- d) mechanical tests.

4.3 Structure of testing equipment

The system diagrams and/or driving conditions of the testing equipment shall comply with the structure specified in each item.

5 Standard conditions

5.1 Standard reference atmosphere

Temperature: 25 °C

Air pressure: 101,3 kPa

NOTE No requirement for relative humidity is given because correction by calculation is generally not possible.

If the parameters to be measured depend on temperature and/or pressure, and the law of dependence is known, the values shall be measured in the conditions specified in 5.3 and, if

necessary, be corrected by calculation to the standard reference atmospheric conditions above.

5.2 Standard atmospheric conditions for reference measurements and tests

If the parameters to be measured depend on temperature, pressure and humidity and the law of dependence is unknown, the atmospheric conditions to be specified shall be selected from the following values, as shown in Table 1.

Table 1 – Standard conditions for reference measurements and tests

Temperature °C	Relative humidity ^a % RH	Air pressure ^a kPa
20 ± 3	45 to 75	86 to 106
25 ± 3		
30 ± 3		
35 ± 3		
^a Inclusive values.		

5.3 Standard atmospheric conditions for measurements and tests

Unless otherwise specified, all tests and measurements shall be carried out under standard atmospheric conditions:

temperature: 15 °C to 35 °C;
relative humidity: 25 % to 85 %, where appropriate;
air pressure: 86 kPa to 106 kPa.

The absolute humidity of the atmosphere shall not exceed 22 g/m³.

5.4 Standard atmospheric conditions for assisted drying

Where assisted drying is required before commencing a series of measurements, the conditions listed below shall be used on the PDP module for at least 2 h, unless otherwise prescribed by the relevant specification:

temperature: (55 ± 3) °C;
relative humidity: not exceeding 20 %;
air pressure: 86 kPa to 106 kPa.

When the specified temperature for the dry heat test is lower than 55 °C, assisted drying shall be carried out at that lower temperature.

5.5 Recovery conditions

The recovery shall be carried out in the conditions specified in IEC 60068-1:2013, 5.3:

temperature: 15 °C to 35 °C;
relative humidity: 25 % to 75 %;
air pressure: 86 kPa to 106 kPa.

5.6 Standard installation conditions

Unless otherwise specified in the relevant specification, stand the PDP module keeping adequate clearance to avoid airflow disturbance. The mounting structure of the PDP module shall be specified in the relevant specification.

5.7 Standard measuring conditions

The standard measuring conditions described in IEC 61988-2-1 shall be applied.

5.8 PDP module state

For the non-operating test, the PDP module shall be either unpacked and turned off, or as otherwise specified in the relevant specification.

For the operating test, the PDP module shall be either in the unpacked, turned off and ready-for-use state, or as otherwise specified in the relevant specification. In the operating test, the module shall be turned on in each test procedure.

5.9 Operating conditions

Full screen: The signal input sets at (15 ± 1) % of white level without gamma correction or an equivalent input level when gamma correction is used.

In case a different signal input is used, it shall be noted in the report.

NOTE The 15 % signal input level is a typical value for video.

6 Measurements

6.1 General

Measurement items are defined in each testing method. An initial measurement is required before each test, and a final measurement is required after the test. If necessary, intermediate measurements can be carried out during the test. If necessary, recovery conditioning defined in each test shall be applied before final and intermediate measurements.

6.2 Measurement item

The following items shall be evaluated on initial, intermediate and final measurements:

- a) visual and optical performance (refer to IEC 61988-2-1);
- b) electrical performance (refer to IEC 61988-2-1);
- c) acoustic noise (refer to IEC 61988-2-5);
- d) mechanical performance.

If additional measurements are carried out, they shall be noted in the report. Data about initial, intermediate and final measurements shall be recorded in the report.

6.3 Initial measurements

Initial measurements shall be carried out as specified under the standard atmospheric conditions for the reference measurements and tests.

6.4 Intermediate measurements

The relevant specification may require functional tests during the conditioning programme. When intermediate measurements are required, the relevant specification shall define the

measuring items and the period(s) during the conditioning, and the results shall be noted in the report.

6.5 Final measurements

Final measurements shall be carried out in the same way as specified for the initial measurements.

6.6 Information to be given in the relevant specification

When any test is included in the relevant specification, at least, the following details shall be given, if applicable:

- a) initial measurements;
- b) final measurements.

Measurement requirements, for example intermediate measurements, etc., are added in each test. Other details, which shall be given in the relevant specification, are described in each clause or subclause.

7 Report

The test conditions shall be noted in the report.

The measurement items, measuring conditions and the results shall be noted in the report.

8 Temperature and humidity testing methods (operation and storage)

8.1 High temperature (storage)

8.1.1 Purpose

The purpose of this test is to evaluate the performance of the PDP module after high temperature storage.

8.1.2 Test conditions

Test Bb of IEC 60068-2-2 shall be applied with the following specific conditions.

NOTE Test Bb: Dry heat for non heat-dissipating specimen with gradual change of temperature.

a) Temperature

The temperature shall be selected from the values given below:

(80 ± 3) °C

(70 ± 3) °C

(60 ± 3) °C

The temperature selected shall be noted in the report.

b) Duration

The duration shall be selected from the values given below:

24 h, 48 h, 72 h, 96 h, 120 h, 240 h and 500 h.

The duration selected shall be noted in the report.

c) Humidity

The absolute humidity of the atmosphere should not exceed 20 g/m³ (corresponding approximately to 50 % relative humidity at 35 °C).

8.1.3 Testing equipment

The specifications of the testing chamber are as follows:

- a) The chamber shall be capable of maintaining the specified temperature in the working space within the tolerances given in IEC 60068-2-2:2007, Clause 4, taking into account the effect of the PDP module under test on chamber conditions. Forced air circulation may be used to maintain homogeneous conditions.
- b) In order to limit radiation problems, the temperature of the walls of the chamber, after temperature stability has been reached, shall not differ by more than 8 % of the specified ambient temperature of the test, expressed in K. This requirement applies to all parts of the chamber walls, and the PDP modules shall be unable to “see” any heating or cooling elements which do not comply with this requirement.
- c) The temperature tolerances given in IEC 60068-2-2:2007, 5.1, are intended to take account of absolute errors in measurements and slow changes of temperature.

8.1.4 Testing procedures

The initial and final measurements shall be applied before and after the test respectively.

The following testing procedure shall be applied:

- a) The chamber shall be at the temperature of the laboratory. The PDP module, while being at the ambient temperature of the laboratory, shall be introduced into the chamber in accordance with 5.6 or as otherwise defined.
- b) The temperature in the chamber shall then be adjusted to the temperature appropriate to the degree of severity and time shall be allowed for the chamber to reach temperature stability. (Temperature stability is defined in IEC 60068-1:2013, 4.8.) The rate of change of temperature in the chamber shall not exceed 3 °C/min, averaged over a period of not more than 5 min. The test temperature shall be measured in accordance with IEC 60068-1:2013, 4.6.
- c) The PDP module shall then be exposed to the high temperature conditions for the duration as specified in the relevant specification. The duration shall be measured from the moment temperature stability has been reached.
- d) At the end of this period, the PDP module shall remain in the chamber and the temperature shall be gradually lowered to a value lying within the limits of standard atmospheric conditions for testing. The rate of change of temperature in the chamber shall not exceed 3 °C/min, averaged over a period of not more than 5 min. At the end of this period, the PDP module shall be subject to the recovery procedure in the chamber or otherwise as appropriate.

8.1.5 Recovery

The following recovery is added to the procedure:

- a) The PDP module shall then remain under standard atmospheric conditions for recovery for a period adequate for the attainment of temperature stability.
- b) If required by the relevant specification, the PDP module shall be measured during the recovery period.

8.1.6 Measurements

The initial and final measurements shall be applied, where the measuring items shall be defined in the relevant specification (see Clause 6). If these are required by the relevant specification, intermediate measurements shall be performed during the procedure c) and d) in 8.1.4.

8.1.7 Information to be given in the relevant specification

In addition to 6.6, the information below shall be given, if applicable:

- a) time duration;
- b) room temperature and PDP module temperature;
- c) if required, intermediate measurements, time and temperature.

8.2 Low temperature (operation)

8.2.1 Purpose

The purpose of this test is to evaluate the performance of the PDP module at low temperature operating conditions.

8.2.2 Test conditions

Test Ad of IEC 60068-2-1 shall be applied with the following specific conditions.

NOTE Test Ad: Cold for heat-dissipating specimens with gradual change of temperature that are powered after initial temperature stabilization.

a) Temperature

The temperature shall be selected from the values given below:

(-20 ± 3) °C

(-10 ± 3) °C

(-5 ± 3) °C

The temperature selected shall be noted in the report.

b) Time before PDP module power-on

After the specified temperature has been reached, the time before the PDP module power-on shall be selected from the values given below:

2 h, 4 h, 8 h, 12 h and 24 h.

The selected time before the PDP module power-on shall be noted in the report.

c) Duration

The duration shall be 4 h as shown in Figure 1 or longer. In case it is longer it shall be noted in the report.

8.2.3 Testing equipment

The specifications of the testing chamber are as follows:

- a) The chamber shall be capable of maintaining the specified temperature in the working space within the tolerances given in IEC 60068-2-1:2013, Clause 4, taking into account the effect of the PDP module under test on chamber conditions. Forced air circulation may be used to maintain homogeneous conditions.
- b) In order to limit radiation problems, the temperature of the walls of the chamber, after temperature stability has been reached, shall not differ by more than 8 % of the specified ambient temperature of the test, expressed in K. This requirement applies to all parts of the chamber walls and the PDP modules shall be unable to “see” any heating or cooling elements which do not comply with this requirement.
- c) The temperature tolerances given in IEC 60068-2-1:2013, 5.1, are intended to take account of absolute errors in measurements and slow changes of temperature. For heat-dissipating PDP modules, the temperature near the PDP module is influenced by the effect of heat dissipation of the PDP module itself and can be different from the values measured in the positions defined in IEC 60068-1.

8.2.4 Testing procedures

The initial and final measurements shall be applied before and after the test respectively.

The following testing procedure shall be applied:

- a) The PDP module shall be introduced into the chamber in the ready-for-use state.
- b) Adjust the temperature in the chamber to the prescribed temperature and maintain at the temperature for the prescribed duration. The temperature and time before the PDP module power-on shall be selected from the values given in 8.2.2.
- c) Operate the PDP module. See 5.9 for the operating conditions.
- d) Measure the PDP module 5 min after the PDP module has been turned on (low temperature, first measurement on Figure 1).
- e) Keep the PDP modules operating for 4 h or longer.
- f) Measure the PDP module (low temperature, second measurement on Figure 1).
- g) Turn off the PDP module and adjust the chamber to the standard atmospheric conditions. The PDP module shall not show condensation.

Figure 1 shows the testing procedure at the low temperature operation mentioned above.

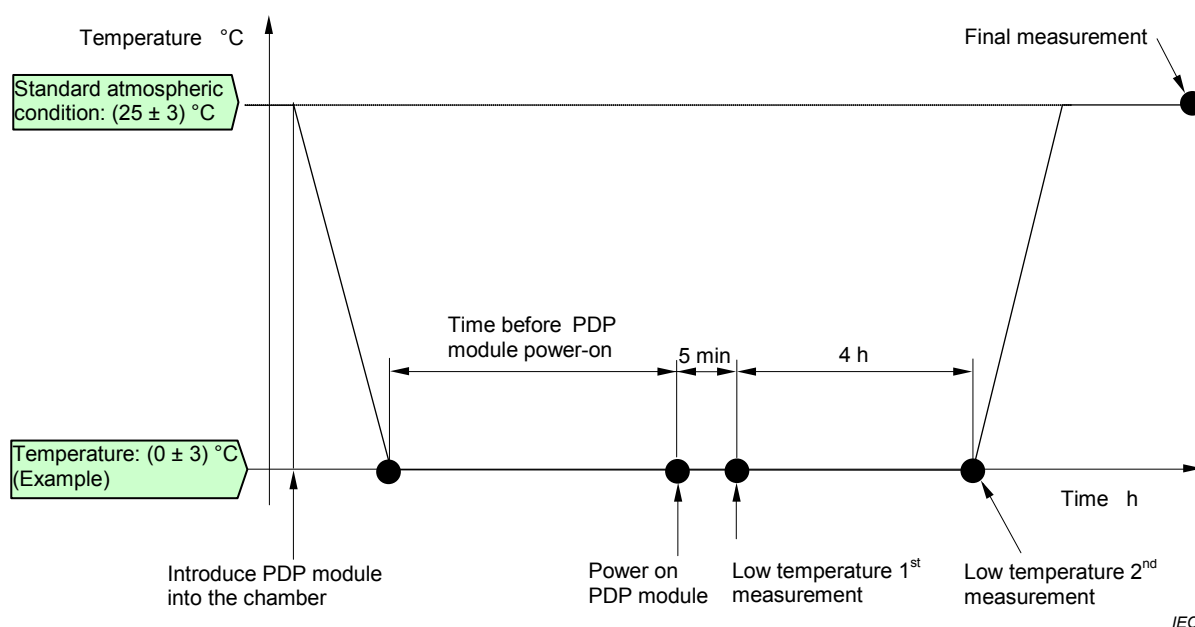


Figure 1 – Testing procedure at low temperature (operation)

8.2.5 Recovery

The following recovery shall be applied:

- a) The droplets of water may be removed.
- b) The PDP module shall then remain under standard atmospheric conditions for recovery for a period until the dew condensation is removed, with a minimum of 5 h.
- c) If required by the relevant specification, the PDP module shall be measured during the recovery period.

8.2.6 Measurements

The initial and final measurements shall be applied, where the measuring items shall be defined in the relevant specification (see Clause 6).

8.2.7 Information to be given in the relevant specification

In addition to 6.6, the information below shall be given, if applicable:

- a) temperature and time before PDP module power-on;
- b) low temperature 1st and 2nd measurements.

8.3 Low temperature (storage)

8.3.1 Purpose

The purpose of this test is to evaluate the performance of the PDP module after low temperature storage.

8.3.2 Test conditions

Test Ab of IEC 60068-2-1 shall be applied with the following specific conditions.

NOTE Test Ab: Cold for non heat-dissipating specimen with gradual change of temperature.

a) Temperature

The temperature shall be selected from the values given below:

(−40 ± 3) °C

(−30 ± 3) °C

(−20 ± 3) °C

(−10 ± 3) °C

(−5 ± 3) °C

(0 ± 3) °C

The temperature selected shall be noted in the report.

b) Duration

The duration shall be selected from the values given below:

24 h, 48 h, 72 h, 96 h, 120 h and 240 h.

The duration selected shall be noted in the report.

8.3.3 Testing equipment

The testing chamber is specified in 8.2.3.

8.3.4 Testing procedures

The initial and final measurements shall be applied before and after the test respectively.

The following testing procedure shall be applied:

- a) The chamber shall be at the temperature of the laboratory. The PDP module, while being at the ambient temperature of the laboratory, shall be introduced into the chamber in accordance with 5.6, or as otherwise specified.
- b) The temperature in the chamber shall then be adjusted to the temperature appropriate to the degree of severity and time shall be allowed for the chamber to reach temperature stability. (Temperature stability is defined in IEC 60068-1:2013, 4.8.) The rate of change of the temperature in the chamber shall not exceed 3 °C/min, averaged over a period of not more than 5 min. The test temperature shall be measured in accordance with IEC 60068-1:2013, 4.6.
- c) The PDP module shall then be exposed to the low temperature conditions for the duration as specified in the relevant specification. The duration shall be measured from the moment temperature stability has been reached.

- d) At the end of this period, the PDP module shall remain in the chamber and the temperature shall be gradually raised to a value lying within the limits of standard atmospheric conditions for testing. The rate of change of the temperature in the chamber shall not exceed 3 °C/min, averaged over a period of not more than 5 min. At the end of this period, the PDP module shall be subject to the recovery procedure in the chamber, or otherwise as appropriate.

8.3.5 Recovery

The following recovery shall be applied:

- a) The droplets of water may be removed.
- b) The PDP module shall then remain under standard atmospheric conditions for recovery for a period until the dew condensation is removed, with a minimum of 5 h.
- c) If required by the relevant specification, the PDP module shall be measured during the recovery period.

8.3.6 Measurements

The initial and final measurements shall be applied, where the measuring items shall be defined in the relevant specification (see Clause 6). If these are required by the relevant specification, intermediate measurements shall be performed during the procedure c) and d) in 8.3.4.

8.3.7 Information to be given in the relevant specification

In addition to 6.6, the information below shall be given, if applicable:

- a) time duration;
- b) room temperature and PDP module temperature;
- c) if required, intermediate measurements, time and temperature.

8.4 Damp heat at steady state (operation)

8.4.1 Purpose

The purpose of this test is to evaluate the performance of the PDP module after exposure to damp heat operating conditions.

8.4.2 Test conditions

This test shall be executed in a chamber or a room in which constant temperature and humidity can be maintained. Water drop condensation should be avoided.

Test Cab of IEC 60068-2-78 shall be applied with the following specific conditions.

NOTE Test Cab: Damp heat, steady state.

a) Temperature and humidity

The temperature and humidity shall be selected from the values given below:

Temperature: (40 ± 3) °C

(50 ± 3) °C

(60 ± 3) °C

Humidity: (85 ± 3) %RH

(93 ± 3) %RH

The temperature and humidity selected shall be noted in the report.

b) Time before PDP module power-on

After the specified temperature and humidity have been reached, the time before the PDP module power-on shall be selected from the values given below:

2 h, 4 h, 8 h, 12 h and 24 h.

The time before the PDP module power-on selected shall be noted in the report.

c) **Duration**

The duration shall be selected from the values given below:

24 h, 48 h, 72 h, 96 h, 120 h, 240 h, 500 h and 1 000 h.

The duration selected shall be noted in the report.

8.4.3 Testing equipment

The testing chamber is specified in IEC 60068-2-78:2012, 4.1.

8.4.4 Testing procedures

The initial and final measurements shall be applied before and after the test respectively.

For testing procedures other than those described above, refer to IEC 60068-2-78:2012, Clause 7.

8.4.5 Recovery

The PDP module shall be placed under the standard atmospheric conditions for assisted drying, or shall remain under standard atmospheric conditions for recovery for a period until the dew condensation is removed, with a minimum of 5 h.

When the PDP module is placed under the standard atmospheric conditions for assisted drying, placing the PDP module under the standard atmospheric conditions (at least 2 h) shall follow.

8.4.6 Measurements

The initial and final measurements shall be applied, where the measuring items shall be defined in the relevant specification (see Clause 6). If these are required by the relevant specification, intermediate measurements shall be performed during the procedure. Intermediate measurements are specified in IEC 60068-2-78:2012, 5.2.

8.4.7 Information to be given in the relevant specification

In addition to 6.6, the information below shall be given, if applicable:

- a) time duration;
- b) room temperature, humidity and PDP module temperature;
- c) if required, intermediate measurements, time and temperature, humidity.

8.5 Damp heat at steady state (storage)

8.5.1 Purpose

The purpose of this test is to evaluate the performance of the PDP module after exposure to damp heat storage.

8.5.2 Test conditions

This test shall be executed in a chamber or a room in which constant temperature and humidity can be maintained. Water drop condensation should be avoided.

Test Cab of IEC 60068-2-78 shall be applied with the following specific conditions.

NOTE Test Cab: Damp heat, steady state.

a) Temperature and humidity

The temperature and humidity shall be selected from the values given below:

Temperature: $(40 \pm 3) ^\circ\text{C}$

$(50 \pm 3) ^\circ\text{C}$

$(60 \pm 3) ^\circ\text{C}$

Humidity: $(85 \pm 3) \% \text{RH}$

$(93 \pm 3) \% \text{RH}$

The temperature and humidity selected shall be noted in the report.

b) Duration

The duration shall be selected from the values given below:

24 h, 48 h, 72 h, 96 h, 120 h and 240 h

The duration selected shall be noted in the report.

8.5.3 Testing equipment

The testing chamber is specified in IEC 60068-2-78:2012, 4.1.

8.5.4 Testing procedures

The initial and final measurements shall be applied before and after the test respectively.

For testing procedures other than described above, refer to IEC 60068-2-78:2012, Clause 7.

8.5.5 Recovery

The PDP module shall be placed under the standard atmospheric conditions for assisted drying, or shall remain under standard atmospheric conditions for recovery for a period until the dew condensation is removed, with a minimum of 5 h.

When the PDP module is placed under the standard atmospheric conditions for assisted drying, placing the PDP module under the standard atmospheric conditions (at least 2 h) shall follow.

8.5.6 Measurements

The initial and final measurements shall be applied, where the measuring items shall be defined in the relevant specification (see Clause 6). If these are required by the relevant specification, intermediate measurements shall be performed during the procedure. Intermediate measurements are specified in IEC 60068-2-78:2012, 5.2

8.5.7 Information to be given in the relevant specification

In addition to 6.6, the information below shall be given if applicable:

- a) time duration;
- b) room temperature, humidity and PDP module temperature;
- c) if required intermediate measurements, time and temperature, humidity.

8.6 Damp heat, cyclic

8.6.1 Purpose

The purpose of this test is to evaluate the performance of the PDP module after a damp heat, cyclic test.

8.6.2 Test conditions

Test Db of IEC 60068-2-30 shall be applied with the following specific conditions.

NOTE Test Db: Damp heat, cyclic.

Test conditions shall be selected from the values given in IEC 60068-2-30:2005, Clause 5.

8.6.3 Testing equipment

The testing chamber is specified in IEC 60068-2-30:2005, Clause 4.

8.6.4 Testing procedure

The initial and final measurements shall be applied before and after the test respectively.

The following testing procedure shall be applied:

- a) The PDP module shall be introduced into the chamber in accordance with 5.6 or as otherwise specified in the relevant specification.
- b) The damp heat conditions (12 h + 12 h cycle) defined in IEC 60068-2-30:2005, Clause 7 shall be applied.
- c) The relevant specification may require functional tests during the conditioning programme.
- d) When intermediate measurements are required, the relevant specification shall define the measuring items and the period(s) during the conditioning, and the results shall be noted in the report.

8.6.5 Recovery

The PDP module shall be placed under the standard atmospheric conditions for assisted drying, or shall remain under standard atmospheric conditions for recovery for a period until the dew condensation is removed, with a minimum of 5 h.

When the PDP module is placed under the standard atmospheric conditions for assisted drying, placing the PDP module under the standard atmospheric conditions (at least 2 h) shall follow.

8.6.6 Measurements

The initial and final measurements shall be applied, where the measuring items shall be defined in the relevant specification (see Clause 6.). If these are required by the relevant specification, intermediate measurements shall be performed during the procedure. Intermediate measurements are specified in IEC 60068-2-30:2005, Clause 8.

8.6.7 Information to be given in the relevant specification

In addition to 6.6, the information below shall be given, if applicable:

- a) time duration;
- b) room temperature, humidity and PDP module temperature;
- c) if required, intermediate measurements, time and temperature, humidity.

8.7 Rapid change of temperature with prescribed time of transition

8.7.1 Purpose

The purpose of this test is to determine the ability of the PDP module to withstand rapid changes of ambient temperature.

8.7.2 Test conditions

Test Na of IEC 60068-2-14 shall be applied with the following specific conditions.

NOTE 1 Test Na: Rapid change of temperature with prescribed time of transition.

a) Temperature

The lower temperature shall be specified in the relevant specification and should be chosen from the temperatures of 8.2.2.

The higher temperature shall be specified in the relevant specification and should be chosen from the temperatures of 8.1.2.

b) Transition time

The transition time should be:

2 min to 3 min;

20 s to 30 s;

less than 10 s.

NOTE 2 The choice of transition time depends on the thermal time constant of the PDP module and the most severe change of temperature conditions it will be subjected to in use.

NOTE 3 The shorter transition times can be applicable by the use of automatic transfer testing equipment.

NOTE 4 In an automatic test equipment of a single chamber system, the transition period means the change over time from cut-off of the control at a test temperature to the commencement of the control at another test temperature. If the chamber temperature is maintained at the temperature of the test place on the way to the transition, such a period is considered as "the PDP module being taken out of the chamber."

8.7.3 Testing equipment

Two separate chambers shall be provided, one for the low temperature and one for the high temperature. Also one single chamber may be used for both low and high temperature exposure.

The air in the chamber shall satisfy the condition described in IEC 60068-2-14:2009, Clause 5.

8.7.4 Testing procedures

The initial and final measurements shall be applied before and after the test respectively.

The PDP module shall be removed from the cold chamber and transferred to the hot chamber.

The transition time consists of the time of removal from one chamber and of the insertion into the second chamber, and possibly the dwell time at the ambient temperature of the laboratory.

8.7.5 Recovery

The recovery shall be carried out in the conditions specified in IEC 60068-1:2013, 5.3, if necessary.

8.7.6 Measurements

Evaluation is performed under the standard measuring conditions as described in 5.7.

- a) The image performance shall be compared to the relevant specification.
- b) The electrical performance shall be compared to the relevant specification.
- c) The visual appearance shall be judged visually.

8.7.7 Information to be given in the relevant specification

In addition to 6.6, the information below shall be given, if applicable:

- a) time duration;
- b) room temperature, humidity and PDP module temperature;
- c) if required, intermediate measurements, time and temperature, humidity.

8.8 Change of temperature with specified rate of change

8.8.1 Purpose

The purpose of this test is to determine the ability of the PDP module to withstand and/or function during changes of ambient temperatures.

8.8.2 Test condition

Test Nb of IEC 60068-2-14 shall be applied with the following specific temperature conditions.

NOTE Test Nb: Change of temperature with specified rate of change.

The lower temperature shall be specified in the relevant specification and should be chosen from the temperatures of 8.2.2.

The higher temperature shall be specified in the relevant specification and should be chosen from the temperatures of 8.1.2.

8.8.3 Testing equipment

The air in the chamber shall satisfy the condition described in IEC 60068-2-14:2009, Clause 5.

8.8.4 Testing procedures

The initial and final measurements shall be applied before and after the test respectively.

The PDP module shall be removed from the cold chamber and transferred to the hot chamber.

The transition time consists of the time of removal from one chamber and of the insertion into the second chamber, and possibly the dwell time at the ambient temperature of the laboratory.

8.8.5 Recovery

The recovery shall be carried out in the conditions specified in IEC 60068-1:2013, 5.3, if necessary.

8.8.6 Measurements

Evaluation is performed under the standard measuring conditions as described in 5.7.

- a) The image performance shall be compared to the relevant specification.
- b) The electrical performance shall be compared to the relevant specification.
- c) The visual appearance shall be judged visually.

8.8.7 Information to be given in the relevant specification

In addition to 6.6, the information below shall be given, if applicable:

- a) time duration;
- b) room temperature, humidity and PDP module temperature;
- c) if required, intermediate measurements, time and temperature, humidity.

9 Air pressure testing methods

9.1 Low air pressure (operation)

9.1.1 Purpose

The purpose of this test is to evaluate the influence of air pressure on the assumption that the PDP module is used at high altitude areas.

9.1.2 Test conditions

Test M of IEC 60068-2-13 shall be applied with the following specific conditions.

NOTE Test M: low air pressure.

a) Air pressure

This test starts at ambient conditions.

Air pressure shall be reduced to 1 kPa/min or slower, starting from atmospheric pressure (86 kPa to 106 kPa) down to P_{op} .

Air pressure, P_{op} , shall be selected from the values given below:

(60 ± 5) kPa

(70 ± 5) kPa

(80 ± 5) kPa

The pressure selected shall be noted in the report.

b) Duration

The duration shall be selected from the values given below:

5 min, 30 min and 2 h.

The duration selected shall be noted in the report.

9.1.3 Testing equipment

The testing chamber is specified in IEC 60068-2-13:1983, Clause 3.

9.1.4 Testing procedures

The initial and final measurements shall be applied before and after the test respectively.

The following testing procedure shall be applied:

- a) The chamber shall be within the temperature range specified by the standard atmospheric conditions for testing. For non-operating tests, the PDP module shall be introduced into the chamber in accordance with 5.6 or as otherwise specified.
- b) The pressure in the chamber shall then be reduced to the value appropriate to the severity.
- c) When operating tests are required, the PDP module shall be switched on or be electrically loaded. Checks shall be made to ascertain whether the PDP module is capable of functioning in accordance with the relevant specification. The PDP module may remain under operating conditions for the specified duration, or be switched off as prescribed in

the relevant specification. For heat-dissipating PDP modules, the relevant specification may require that the PDP module is switched on and adequate time allowed for it to reach thermal stability before or after the air pressure is reduced and functional tests and/or measurements are made.

- d) The air pressure shall be maintained for the specified duration.
- e) The air pressure shall be restored to the standard atmospheric conditions at the end of the tests and measurements. Even if required by the relevant specification, the rate of change of air pressure shall not exceed 10 kPa/min.

9.1.5 Recovery

The following recovery shall be applied:

- a) The droplets of water may be removed.
- b) The PDP module shall then remain under standard atmospheric conditions for recovery for a period until the dew condensation is removed.

9.1.6 Measurements

The initial and final measurements shall be applied, where the measuring items shall be defined in the relevant specification (see Clause 6). If required by the relevant specification, intermediate measurements shall be performed during the procedure.

9.1.7 Information to be given in the relevant specification

In addition to 6.6, the information below shall be given, if applicable:

- a) time duration;
- b) room temperature, air pressure;
- c) if required intermediate measurements, time and temperature, air pressure.

9.2 Low air pressure (storage)

9.2.1 Purpose

The purpose of this test is to evaluate the influence of air pressure on the assumption that the PDP module is stored or transported at high altitude areas.

9.2.2 Test conditions

Test M of IEC 60068-2-13 shall be applied with the following specific conditions.

NOTE Test M: low air pressure.

a) Air pressure

Air pressure shall be reduced to 4 kPa/min or slower, starting from atmospheric pressure (86 kPa to 106 kPa) down to P_{st} .

Air pressure, P_{st} , shall be selected from the values given below:

(30 ± 5) kPa

(40 ± 5) kPa

(50 ± 5) kPa

The pressure selected shall be noted in the report.

b) Duration

The duration shall be selected from the values given below:

5 min, 30 min and 2 h.

The duration selected shall be noted in the report.

9.2.3 Testing equipment

Testing chamber is specified in IEC 60068-2-13:1983, Clause 3.

9.2.4 Testing procedures

The initial and final measurements shall be applied before and after the test respectively.

The following testing procedure shall be applied:

- a) The chamber shall be within the temperature range specified by the standard atmospheric conditions for testing. For non-operating tests, the PDP module shall be introduced into the chamber in accordance with 5.6 or as otherwise specified.
- b) The pressure in the chamber shall then be reduced to the value appropriate to the severity.
- c) When operating tests are required, the PDP module shall be switched on or be electrically loaded. Checks shall be made to ascertain whether the PDP module is capable of functioning in accordance with the relevant specification. The PDP module may remain under operating conditions for the specified duration, or be switched off as prescribed in the relevant specification. For heat-dissipating PDP modules, the relevant specification may require that the PDP module is switched on and adequate time allowed for it to reach thermal stability before or after the air pressure is reduced and functional tests and/or measurements are made.
- d) The air pressure shall be maintained for the specified duration.
- e) The air pressure shall be restored to the standard atmospheric conditions at the end of the tests and measurements. Even if required by the relevant specification, the rate of change of air pressure shall not exceed 10 kPa/min.

9.2.5 Recovery

The following recovery shall be applied:

- a) The droplets of water may be removed.
- b) The PDP module shall then remain under standard atmospheric conditions for recovery for a period until the dew condensation is removed.

9.2.6 Measurements

The initial and final measurements shall be applied, where the measuring items shall be defined in the relevant specification (see Clause 6). If required by the relevant specification, intermediate measurements shall be performed during the procedure.

9.2.7 Information to be given in the relevant specification

In addition to 6.6, the information below shall be given, if applicable:

- a) time duration;
- b) room temperature, air pressure;
- c) if required intermediate measurements, time and temperature, air pressure.

10 Mechanical testing methods

10.1 Vibration (sinusoidal)

10.1.1 Purpose

The purpose of this test is to evaluate the mechanical endurance in a vibration environment.

10.1.2 Equipment

The equipment, including a vibration table and mounting jig, shall be capable of maintaining the test conditions specified in 10.1.3. It should not resonate in the vibration frequency range.

10.1.3 Test conditions

IEC 60068-2-6 and IEC 61747-5:1998, 2.3, shall be applied with the following specific conditions. In case of contradiction between these standards, IEC 61747-5:1998, 2.3, shall govern.

- a) Vibration frequency: continuously sweep from (10 ± 1) Hz to (55 ± 1) Hz, sweep back to (10 ± 1) Hz and repeat.
- b) Duration: 20 min each axis.
- c) Vibration axes: Figure 2 shows the axes of the PDP module.
- d) Acceleration: The acceleration shall be selected from the values given below:
10,0 m/s², 10,5 m/s², 11,0 m/s², 11,5 m/s², 12,0 m/s², 12,5 m/s² and 13,0 m/s².
- e) The acceleration selected shall be noted in the report.
- f) Sweep rate: 1 octave/min with a tolerance of ± 10 %.
- g) The PDP module shall be mounted on a testing table.

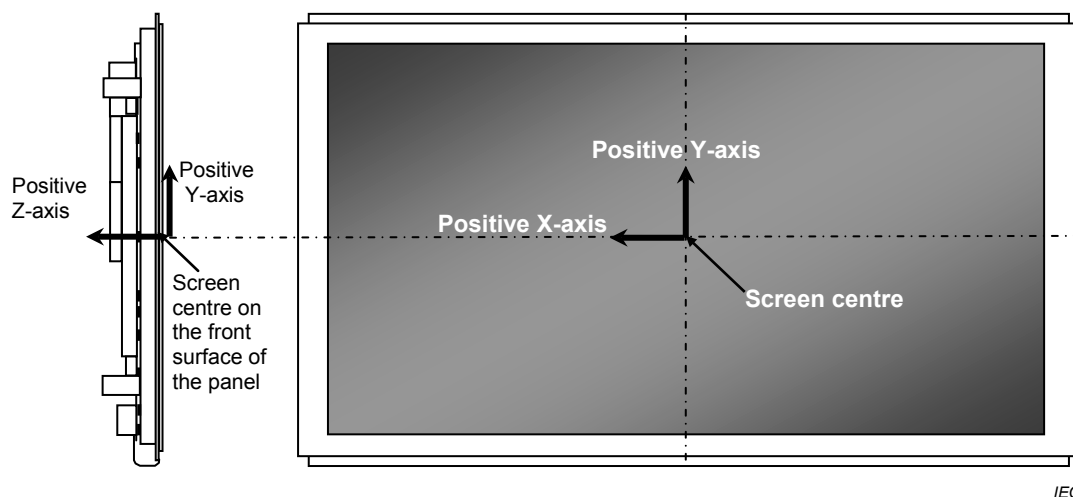


Figure 2 – Configuration of PDP module directions

10.2 Shock

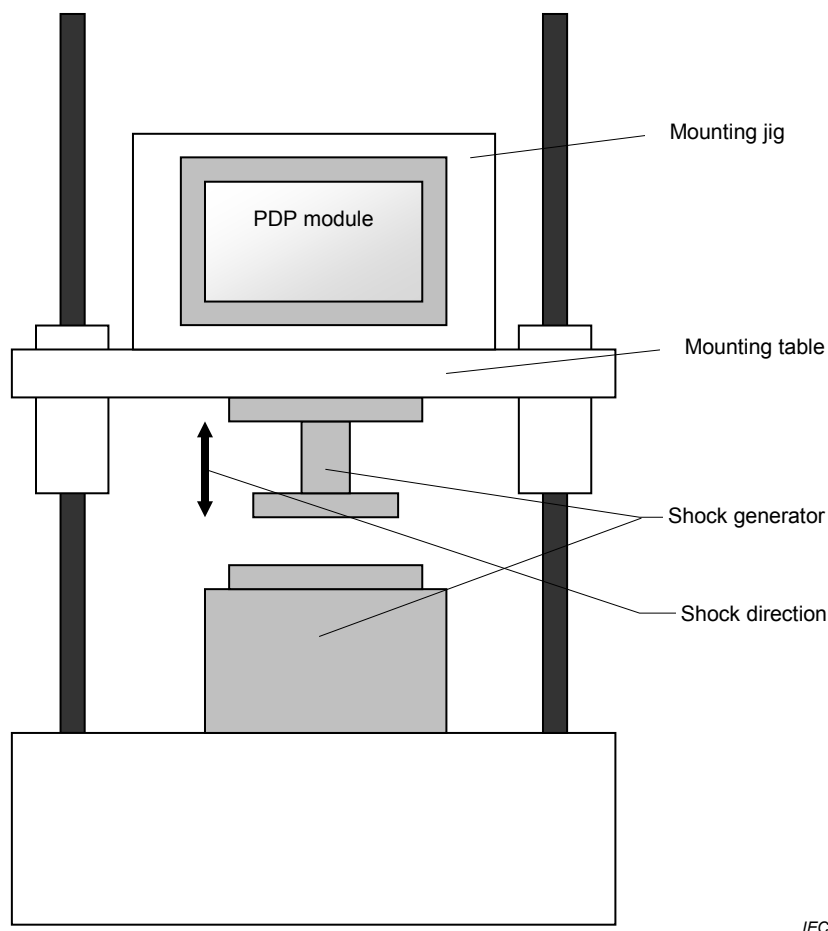
10.2.1 Purpose

The purpose of this test is to evaluate the mechanical endurance after mechanical shocks.

10.2.2 Equipment

The equipment, including a shock generator and mounting jig, shall be capable of maintaining the test conditions specified in 10.2.3. It should not produce unnecessary vibration.

Figure 3 shows an example of shock testing equipment. The PDP module shall be clamped during the test.



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Figure 3 – Example of PDP shock testing equipment

10.2.3 Test conditions

IEC 60068-2-27 and IEC 61747-5:1998, 2.4, shall be applied with the following specific conditions.

In case of contradiction between these standards, IEC 61747-5:1998, 2.4, shall govern.

The waveform to be used for acceleration is half sine as shown in IEC 60068-2-27:2008, Figure 2.

The peak value of acceleration for each direction shall be selected from Table 2. Refer to Figure 2 for X, Y and Z.

Table 2 – Peak value and duration

	X m/s ²	Y m/s ²	Z m/s ²	Duration ms
Condition 1	100	100	30	30
Condition 2	150	150	50	11
NOTE Gravitational acceleration at the earth's surface is approximately 10 m/s ² .				

A PDP module with a screen diagonal equal to or larger than 1,25 m (50-inch type) is recommended to be tested at condition 1.

A PDP module with a screen diagonal smaller than 1,25 m (50-inch type) is recommended to be tested at condition 2.

The condition selected shall be noted in the report.

Application of shocks:

The PDP module shall be subjected to three successive shocks in both positive and negative directions of three mutually perpendicular axes chosen so that faults are most likely to be revealed, i.e. a total of 18 shocks.

10.3 Transportation drop

Refer to ISO 2248 and ISO 4180.

10.4 Toppling

Refer to ISO 10531.

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